

TrustVote: Online Voting using BlockChain

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Abstract - The purpose of this system is to provide a secure, transparent, and immutable platform for conducting elections. The system employs the use of smart contracts on the Ethereum blockchain to automate the voting process and ensure that all votes are counted accurately. The system was developed using Solidity, a programming language for smart contracts, and Truffle, a development framework for Ethereum. The system's frontend was developed using HTML, CSS and JavaScript. Nodejs, Expressjs and MongoDB for system's backend. The system's security was ensured through the use of encryption techniques and private keys to secure users' identities and their voting records. The system's architecture allows for the registration of voters, the creation of ballots, the casting of votes, and the tallying of results, all on the blockchain. The system's decentralized nature ensures that votes cannot be tampered with, and the results are transparent and available for all participants to verify. This project report concludes that a decentralized voting system using blockchain technology has the potential to revolutionize the way elections are conducted, by providing a secure, transparent, and immutable platform that all participants can trust. The system can be used in various settings, including governmental and non-governmental elections, corporate boardroom elections, and more.

Key Words: Ethereum, Blockchain voting, Decentralization, Smart-contract

1. INTRODUCTION

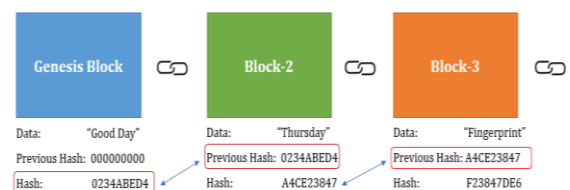
Democratic voting is such an important and significant event in every country, and the current way we vote usually involves either ballot papers or EVMs. While these methods are in place, they do come with some challenges, like transparency issues, lower voter participation, vote tampering, a lack of trust in the election authority, problems with unique voter IDs, delays in announcing results, and, most importantly, security concerns. When we think about implementing a digital voting system, the security of digital voting really stands out as a primary concern. Given the enormous decisions involved, we want to be absolutely sure that the system can protect data and guard against possible attacks. One potential solution to these security challenges could be the use of blockchain technology. Blockchain technology opens up countless possibilities. It's a distributed ledger technology that enables digital assets to be exchanged in a peer-to-peer decentralized network. This advancement in distributed ledger technology is truly exciting! A block is essentially a collection of all the transactions that have taken

place. Blockchain comes with fantastic features like immutability, decentralization, security, transparency, and anonymity. Combining blockchain with smart contracts presents a wonderful opportunity for creating a safer, more secure, and transparent e-voting system. In this paper, we've worked on and tested a sample e-voting application as a smart contract for the Ethereum network, utilizing blockchain technology through wallets and the Solidity programming language.

1.1 Blockchain Technology

This represents a decentralized network comprising interconnected nodes. Each node is allocated a copy of the distributed ledger, which encompasses a comprehensive record of all transactions executed by the network. Every processed transaction produces a unique hash. The generated hash is contingent not solely on the current transaction but also on the hash of the preceding transaction.

Consequently, any minor alteration to the data will influence the hash of the transaction. The transaction is inscribed into the block. This configuration enables users to maintain their autonomy while engaging with the system. A fundamental examination of Blockchain indicates that it possesses the capability to enhance the security and dependability of the voting process.



2. OBJECTIVE

The primary objective of this project is to design and implement a decentralized voting system that leverages blockchain technology to enhance the integrity, security, and transparency of the electoral process. Specific objectives include: (1) developing a user-friendly web application that allows voters to register, cast votes, and view results; (2) utilizing smart contracts on the Ethereum blockchain to automate voting processes and ensure accurate vote counting; (3) implementing robust security measures, including encryption and user authentication via MetaMask, to protect voter identities and data; and (4) evaluating the

system's effectiveness in preventing fraud and ensuring verifiable election outcomes. Ultimately, this project aims to demonstrate the potential of blockchain technology to revolutionize traditional voting systems.

3. LITERATURE SURVEY

Bhalerao et al. [1] proposed a secure digital voting system using blockchain technology. The study emphasizes the system's architecture, security features, and implementation details. While primarily focused on the technical aspects, the discussion on scalability and user accessibility is limited. This study further builds on the concept by addressing additional aspects such as scalability and practical deployment in large-scale elections.

Pashine, Naive, and Kelapure [2] have come up with a delightful Android platform for online voting. This application streamlines the lengthy voting process while ensuring the security of voters and making the experience more comfortable. Voters can easily cast their votes for candidates right from their hometown without needing to go to a polling booth. Plus, it even features gesture recognition, although the challenge of ensuring authentication remains on the Android platform. This user-friendly application is thoughtfully divided into three panels: the Admin Panel, Candidate Panel, and Voter Panel.

Khasawneh [3] has introduced an innovative E-Voting System with biometric security that offers a two-sided solution, addressing both the server and user side. After casting their vote, the system generates a hard copy for the voter, along with a unique number. This unique number, along with the voter's name and identification number, is kept secure. All information is stored in a special, secure box, which is used for verifying the vote before it is added to the final database. The printed side copy features a unique barcode that can be easily scanned; one copy is randomly selected for testing. This two-sided process ensures verification and accuracy for the entire system.

Kiayias et al. [4] implemented a blockchain-based e-voting system, focusing on privacy, verifiability, and resistance to tampering. However, the study lacked practical deployment analysis and did not consider large-scale election scenarios. The current study compares these implementation strategies and enhances the security protocols for better real-world applicability.

Aravind, Gokul Raj, and Mohanraj [5] proposed a system incorporating fingerprint authentication for identity verification, highlighting its speed and reliability over other biometric data. The system utilizes blockchain data storage and homomorphic encryption. It features two types of users: the Election Officer and the Booth Manager, with the latter managing voter functionalities during polling. The paper notes that no full implementation of blockchain-based e-voting has been achieved for a national election. While the system aims to reduce costs associated with traditional voting methods, the complexity of implementation presents significant challenges. Additionally, the system seeks to

decentralize the voting process, potentially enhancing security and transparency.

4. PROBLEM STATEMENT

Traditional voting systems often suffer from issues such as voter fraud, lack of transparency, and slow results, which can undermine public trust and participation. A blockchain-based e-voting system offers a solution by providing a secure, transparent, and tamper-proof platform for elections. This project aims to develop a blockchain e-voting system to enhance the integrity and efficiency of the voting process.

5. EXISTING SYSTEM

Electronic voting (often called e-voting) is all about using electronic methods to support or manage the process of casting and counting votes in a friendly and efficient way. Depending on how it's set up, e-voting can involve stand-alone electronic voting machines (lovingly known as EVMs) or computers that are connected to the Internet. It can include a variety of online services, from simply transmitting tabulated results to fully functional online voting accessible through common household devices. An effective e-voting system should be able to handle most of these tasks while adhering to a cheerful set of standards laid out by regulatory bodies, ensuring it's cost-effective, scalable, and environmentally friendly. While much of the current discussion focuses on security, accuracy, integrity, speed, privacy, and auditability, it's important to note that existing systems can fall short when it comes to protecting against attacks at certain levels.

Some of the challenges of the current system:

- Centralized architecture. –
- Vulnerable to attacks. –
- Lacks trustworthiness. –
- Vote casting process is not transparent.

6. PROPOSED SYSTEM

The proposed system is a decentralized online voting platform using blockchain technology for security, transparency, and immutability. Built on Ethereum, it uses smart contracts to automate and validate the voting process, ensuring accurate vote counting.

Key features include:

1. **Voter Registration:** Voters register with a unique identifier and are authenticated securely. This data is stored on the blockchain.
2. **Voting Process:** Authenticated users cast their vote, which is recorded on the blockchain, ensuring transparency and preventing tampering.
3. **Smart Contracts:** Automate vote casting, ballot creation, and result tallying, ensuring trust and transparency.
4. **Admin Panel:** Admins manage voter registrations, add candidates, and view results, without influencing the vote count.

5. **Security:** Encryption and private keys ensure voter confidentiality, with votes stored immutably on the blockchain.
6. **Real-time Results:** Election results are available in real-time on the dashboard, verifiable by all participants.

This system ensures secure, transparent elections with no central authority capable of manipulating the results.

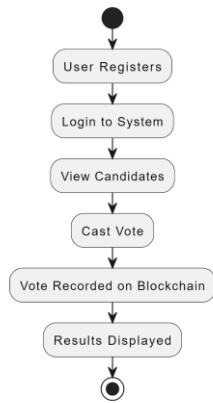


Fig. Process Flow

6.1. Technology Used:

1. **Blockchain Technology:**
 - **Ethereum:** Provides a decentralized platform for secure voting.
 - **Ganache:** Used for local blockchain development and testing of smart contracts.
2. **Smart Contracts:**
 - Developed with **Solidity** to automate voting processes and enforce election rules.
3. **Web Technologies:**
 - **Frontend:** Built using **HTML**, **CSS**, and **JavaScript** for an interactive user interface.
 - **Backend:** Utilizes **Node.js** and **Express.js** for server-side operations.
4. **Database:**
 - **MongoDB:** Stores user information and voting records securely.
5. **User Authentication:**
 - **MetaMask:** Enables secure user authentication and interaction with the Ethereum blockchain.
6. **Version Control:**
 - **Git** and **GitHub:** For version control and collaboration.
7. **Testing Tools:**
 - **Postman:** Used for API testing to ensure functionality.

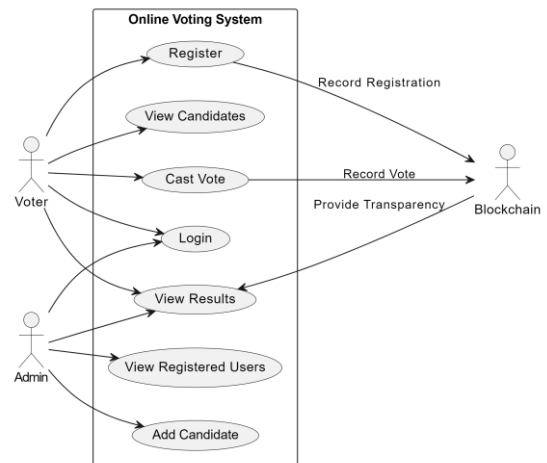


Fig. Use Case Diagram

6.2. Significance of the Proposed Method

1. **Enhanced Security:**
 - Utilizes blockchain technology to provide a secure and tamper-proof voting environment, significantly reducing the risk of fraud and unauthorized access.
2. **Transparency:**
 - All transactions are recorded on a public ledger, allowing for real-time verification of votes and results by all stakeholders, fostering trust in the electoral process.
3. **Decentralization:**
 - Eliminates the need for a central authority, reducing the potential for bias and manipulation. The decentralized nature of the system ensures that no single entity has control over the voting data.
4. **Automation through Smart Contracts:**
 - Automates the voting process, including ballot creation and result tallying, which minimizes human error and increases efficiency.
5. **User Empowerment:**
 - Provides users with a straightforward and secure interface for casting votes, ensuring accessibility for all eligible voters.
6. **Cost-Effective:**
 - Reduces administrative costs associated with traditional voting systems by minimizing paperwork and streamlining the electoral process.
7. **Scalability:**
 - The modular design of the system allows for easy scalability, enabling its application in various electoral contexts, including governmental, corporate, and organizational elections.

7. SYSTEM ARCHITECTURE

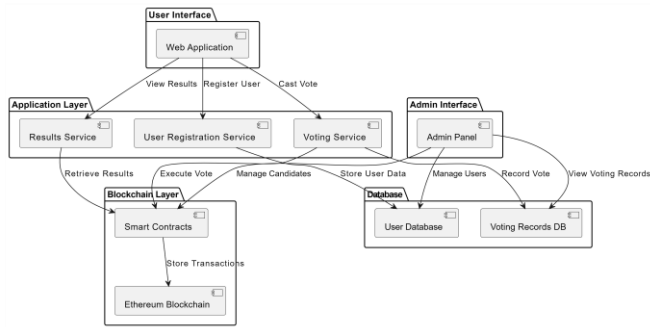


Fig. System Architecture

1. User Interface:

- **Web Application:** The primary interface through which users interact with the voting system. Users can register, cast votes, and view results.

2. Application Layer:

- **Voting Service:** Handles the logic for casting votes and interacting with the blockchain.
- **User Registration Service:** Manages user registration, storing user data securely.
- **Results Service:** Retrieves and displays the results of the election.

3. Blockchain Layer:

- **Ethereum Blockchain:** The underlying decentralized ledger that stores all transactions, ensuring security and immutability.
- **Smart Contracts:** Automated contracts that execute voting actions, enforce rules, and maintain integrity of the voting process.

4. Database:

- **User Database:** Stores user information and credentials.
- **Voting Records Database:** Keeps a record of all votes cast, ensuring transparency and traceability.

5. Admin Interface:

- **Admin Panel:** A secure interface for administrators to manage users, oversee voting records, and add candidates, ensuring smooth operations and monitoring.

8. Modules and Description

1. User Registration Module:

- **Description:** Facilitates user registration by collecting essential information (e.g., Aadhaar number, password) and verifying user identity. The data is securely stored on the blockchain.

2. Voting Module:

- **Description:** Enables authenticated users to cast their votes for selected candidates. Each vote is securely recorded on the Ethereum blockchain, ensuring transparency and immutability.

3. Smart Contract Module:

- **Description:** Automates the voting process, including the creation of ballots and the execution of voting actions. Smart contracts ensure that the

rules of the election are enforced and that votes are counted accurately.

4. Results Module:

- **Description:** Retrieves and displays real-time election results. This module allows users to view the outcomes of the voting process directly from the blockchain, ensuring transparency.

5. Admin Management Module:

- **Description:** Provides a secure interface for administrators to manage user registrations, oversee candidates, and monitor the voting process. Admins can add candidates and view voting records without altering the integrity of the votes.

6. Security Module:

- **Description:** Implements encryption techniques and uses private keys to protect user identities and voting records. Ensures that only authorized users can access sensitive information.

7. Notification Module:

- **Description:** Sends email notifications to users upon successful registration and vote casting, confirming their actions and providing transparency in the voting process.

BLOCKCHAIN-VOTE Register / Login / About / Results / Admin Panel

Create an account

First Name	Last Name
Shrinivas	Kathure
Aadhaar Number	Mobile No.
413897271457	951180405
Email	
shrinivas.22110726@vit.ac.in	
Password	

Already have an account? Click here to login

Your Account: 0xad5a20f6588a98ccceff10aabf3aec07c808a225

Fig. Sign in

BLOCKCHAIN-VOTE Register / Login / About / Results / Admin Panel

Login to an existing account

Aadhaar Number
shrinivas.22110726@vit.ac.in
Password

Your Account: 0xad5a20f6588a98ccceff10aabf3aec07c808a225

Fig. Login in

BLOCKCHAIN-VOTE Register / Login / About / Vote / Results / Admin Panel

Please cast your vote

#	Name	Party Name	Votes
1	Narendra Modi	Bharatiya Janata Party	0
2	Rahul Gandhi	Indian National Congress	0
3	Arvind Kejriwal	Aam Aadmi party	0
4	Mamata Banerjee	All India Trinamool Congress	0

Select Candidate

Your Account: 0xad5a20f6588a98ccceff10aabf3aec07c808a225

Fig. Vote Page

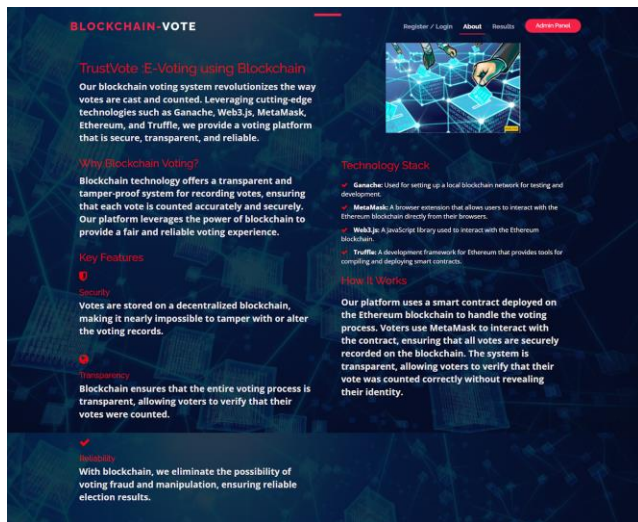


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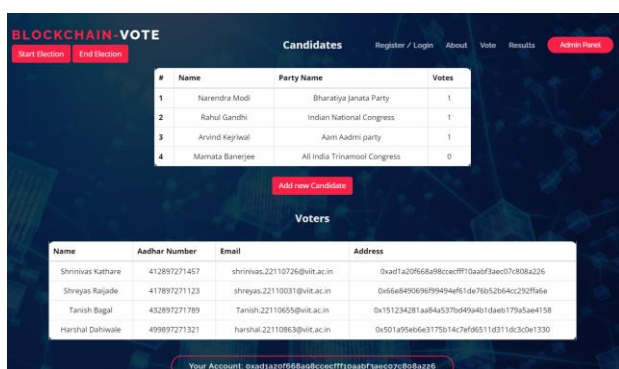


Fig. Admin Panel

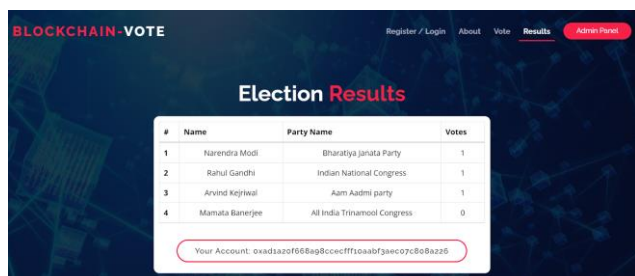


Fig. Result

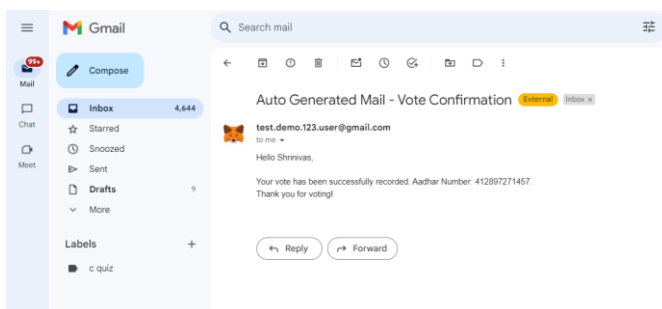


Fig. Vote Confirmation Mail

9. CONCLUSION AND FUTURE SCOPE

The creation of a decentralized online voting system via Blockchain, especially Ethereum, presents transformative potential for electoral processes. This study illustrates enhanced transparency and security through the removal of intermediaries and the application of smart contracts to safeguard result integrity. While our mock voting data indicates practical feasibility, additional real-world trials are necessary for thorough performance evaluation. The analysis of the system's security, scalability, and usability has identified both its advantages and areas needing enhancement. Future improvements, such as incorporating biometric authentication, will strengthen security and broaden accessibility for diverse voter demographics. Ultimately, this research signifies a pivotal advancement towards adopting Blockchain-based decentralized voting systems as reliable substitutes for conventional methods, thus preserving the integrity of democratic processes in a rapidly digitizing context.

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